Kinetic energy and net work

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect net work to a change in kinetic energy.

SUCCESS CRITERIA - I can...

1. Calculate kinetic energy with speed squared.
2. Use net work rather than one force's work.
3. Connect the sign of work to a speed change.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use "kinetic energy" and "work" to explain your model or result. Sentence starters are optional.

kinetic energy: energy associated with motion

work: energy transferred by a force along a displacement

initial: at the beginning of the interval; subscript i

01 Bring back an earlier idea

At fixed momentum magnitude, mass becomes 3 times as large. Derive the kinetic-energy multiplier using $K=p^2/(2m)$.

Word help: momentum = mass times velocity, including direction

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $K = \frac{1}{2}mv^2$; $W_{net} = K_f - K_i$.
i = initial; f = final. Positive net work increases kinetic energy.

Worked reasoning: A 2 kg cart at 2 m/s has K=4 J. Adding 5 J of net work gives final K=9 J, so final speed is 3 m/s.

Your turn: Draw before-and-after energy bars and label the transfer between them.

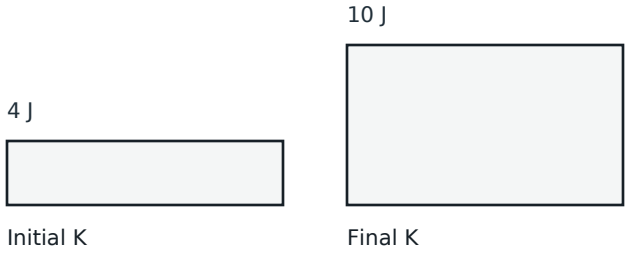
Sentence starter: The ... in my model represents ... . Its label shows ... .

Marzano check - circle: ① ② ③

03 Represent and interpret

Energy bars show kinetic energy changing from 7 J to 10 J. **Find** net work and describe the speed change for constant mass.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Bar height represents energy. Read the value and unit above each bar.

Marzano check - circle: ① ② ③

Kinetic energy and net work

antocciphysics

Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Solve and justify

A 4 kg cart moves at 2 m/s. Net work is 24 J.
Find final kinetic energy and speed.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3

05 Analyze the reasoning

A student adds the work of a pull but ignores friction. Why might their predicted speed be too high?

Sentence starter: I predict ... . My model supports this because ... .

Marzano check - circle: 1 2 3

06 Choose a method

Two equal 4 kg carts collide elastically along one line. A starts at +6 m/s; B starts at rest.
Find both final velocities under the ideal model.

Word help: ideal = using stated simplifying conditions

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3

07 Exit check

A 2 kg object slows from 6 to 2 m/s.
Find net work.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A 3 kg object starts at 4 m/s and receives -18 J of net work.
Find final speed. What is the most negative net work possible before it first stops?

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: 1 2 3